This assignment will not be graded and is only for practice.

5.1 Level 1:

1) Choose the set of correct options **_1 point_**

- [ ] If A is a real 3×3 matrix, then $det(A) = det(A^T)$.
- [ ] If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i-th row and j-th column of A is 4×4.
- [ ] If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i-th row and j-th column of A is 3×3.
- [ ] If A is a real 3×3 matrix, then the orders of all possible square sub matrices of A are $1 \times 1, 2 \times 2,$ and 3×3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If A is a real 3×3 matrix, then $det(A) = det(A^T)$.
If $A = [a_{ij}]$ is a real 4×4 matrix, then the order of the sub matrix obtained by deleting the i-th row and j-th column of A is 3×3.
If A is a real 3×3 matrix, then the orders of all possible square sub matrices of A are $1 \times 1, 2 \times 2,$ and 3×3.

2) Let $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix}$ be a 3×3 matrix. Which of the following is(are) correct? **_1 point_**

[Hint: Determinant can be calculated by expanding with respect to different rows or columns.]

- [ ] $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$
- [ ] $det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$
- [ ] $det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$

- [ ] $det(A) = 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) - 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$
$det(A) = 3 \times det(\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix}) + 2 \times det(\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix})$
$det(A) = -2 \times det(\begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}) + 3 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}) - 2 \times det(\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix})$

3) If all the elements of a 3×3 real matrix A are the same, then which of the following is (are) correct? ***1 point***

- [ ] Determinant of matrix A is 0.
- [ ] Determinant of matrix A cannot be determined from the given information.
- [ ] Determinant of matrix A will be the sum of the elements of a row.
- [ ] Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.
- [ ] Determinant of matrix $A + A^T$ cannot be determined from the given information, where A^T denotes the transpose of A.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Determinant of matrix A is 0.
Determinant of matrix $A + A^T$ is 0, where A^T denotes the transpose of A.

4) Choose the set of correct options ***1 point***

- [ ] If two rows are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
- [ ] If two columns are the same in a 3×3 real matrix, then the determinant of that matrix is zero.

- [ ] If one row is a non-zero multiple of another row in a 3×3 real matrix, then the determinant of that matrix is not zero.
- [ ] If one column is a non-zero multiple of another column in a 3×3 real matrix, then the determinant of that matrix is zero.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If two rows are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
If two columns are the same in a 3×3 real matrix, then the determinant of that matrix is zero.
If one column is a non-zero multiple of another column in a 3×3 real matrix, then the determinant of that matrix is zero.

5) Let $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ be a matrix of order 4×4. Find $det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

5.2 Level 2:

6) Let A, B, and C be 3×3 real matrices. Which of the following options is (are) correct? ***1 point***

- [ ] $det(ABC) = det(A)det(B)det(C)$
- [ ] $det(A^3) = det(A)^3$
- [ ] $det(A + B + C) = det(A) + det(B) + det(C)$
- [ ] $det(AB^T) = det(A)\ det(B)$, where B^T denotes the transpose of B.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det(ABC) = det(A)det(B)det(C)$
$det(A^3) = det(A)^3$
$det(AB^T) = det(A)\ det(B)$, where B^T denotes the transpose of B.

7) Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i > j \\ j & \text{if } i \leq j \end{cases}$

1 point

Choose the set of correct options

[Hint: Write down matrix A first.]

- [ ] A is an upper triangular matrix.
- [ ] A is a lower triangular matrix.
- [ ] $det(A) = 6$
- [ ] $det(A) = 3$

No, the answer is incorrect.
Score: 0

Accepted Answers:

A is an upper triangular matrix.
$det(A) = 6$

8) Let $A = [\delta_{ij}]$ be a matrix of order 3×3 such that $\delta_{ij} = \begin{cases} 0 & \text{if } i < j \\ 2 & \text{if } i \geq j \end{cases}$

1 point

Choose the set of correct options

- [ ] A is an upper triangular matrix.
- [ ] A is a lower triangular matrix.
- [ ] $det(A) = 2$
- [ ] $det(A) = 8$

No, the answer is incorrect.

Score: 0

Accepted Answers:

A is a lower triangular matrix.

$det(A) = 8$

9) Suppose $A = \begin{bmatrix} 4 & -1 & 3 \\ 2 & 0 & 1 \\ 3 & -2 & 0 \end{bmatrix}$ and C_{ij} is the (i, j)-th cofactor of the matrix A.

Let B be 3 × 3 matrix where (i, j)-th entry is equal to C_{ij}. Find $det(B)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 49

1 point

10) Suppose A and B are two 3 × 3 matrices such that $det(A)$ = 4 and $B = 3A$. Find the value of $\sqrt[3]{det(A^2B)}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 12

1 point

Your score is: 0/10